7 Storing your objects: S3

This chapter covers

- Transferring files to S3 using the terminal
- Integrating S3 into your applications with SDKs
- Archiving data at low costs
- Hosting a static website with S3
- Diving into the internals of the S3 object store

Storing data comes with two challenges: dealing with ever-increasing volumes of data and ensuring durability. Solving the challenges is hard, or even impossible, if using disks connected to a single machine. For this reason, this chapter covers a revolutionary approach: a distributed data store consisting of a large number of machines connected over a network. This way, you can store nearly unlimited amounts of data by adding additional machines to the distributed data store. And because your data is always stored on more than one machine, you dramatically reduce the risk of losing that data.

You will learn about how to store images, videos, documents, executables, or any other kind of data on Amazon S3 in this chapter. Amazon S3 is a simple-to-use, fully managed distributed data store provided by AWS. Data is managed as objects, so the storage system is called an *object store*. We will show you how to use S3 to back up your data, how to archive data at low cost, and how to integrate S3 into your own application for storing user-generated content, as well as how to host static websites on S3.

Not all examples are covered by the Free Tier

The examples in this chapter are not all covered by the Free Tier. A warning message appears when an example incurs costs. As for the other examples, as long as you follow the instructions and don't run them longer than a few days, you won't pay anything.